

10ISIBSTAT9

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Problem

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function having the following property: For any two points A and B in $\mathbb{R}^2$, the distance between A and B is the same as the distance between the points $f(A)$ and $f(B)$.

Denote the unique straight line passing through A and B by $l(A, B)$

(a) Suppose that C, D are two fixed points in $\mathbb{R}^2$. If X is a point on the line $l(C, D)$, then show that $f(X)$ is a point on the line $l(f(C), f(D))$.

(b) Consider two more point E and F in $\mathbb{R}^2$ and suppose that $l(E, F)$ intersects $l(C, D)$ at an angle α . Show that $l(f(C), f(D))$ intersects $l(f(E), f(F))$ at an angle α . What happens if the two lines $l(C, D)$ and $l(E, F)$ do not intersect? Justify your answer.

Solution. Let $d(A, B)$ denote the distance between the points A and B . We shall denote $f(A) = A'$ for the ease of notation

As a result, according to the question, we have $d(A, B) = d(A', B')$.

Lemma 1. For any 3 points in the plane, $d(A, C) + d(C, B) \geq d(A, B)$ with equality iff C lies on the segment AB .

(a) Since $X \in l(C, D)$ (wolog say on the segment CD) then $d(C, X) + d(X, D) = d(C, D) \implies d(C', X') + d(X', D') = d(C', D') \implies X' \in l(C', D')$ (to be specific X' lies on the segment $C'D'$). If X was outside the segment $C'D'$ (wolog say closer to D') then we can switch the roles of X and D and would get the same result.

(b) Say $l(C, D)$ intersects $l(E, F)$ at X ($\neq C, \neq E$ wolog). So, $X' \in l(C', D')$ and $X' \in l(E', F')$.

Since $X \neq C \implies d(X, C) \neq 0 \implies d(X', C') \neq 0 \implies X' \neq C'$. With the same argument, we have $X' \neq E'$.

Claim 1 — $\triangle EXC \cong \triangle E'X'C'$.

Proof. We know,

$$\begin{aligned} d(E, X) &= d(E', X') \\ d(X, C) &= d(X', C') \\ d(C, E) &= d(C', E') \end{aligned}$$

As a result, due to the S-S-S criterion, we are done! $\square$

So, $\angle EXC = \angle E'X'C'$ and as a result $\ell(C', D')$ intersect $\ell(E', F')$ at an angle α as well.

Now if $\ell(C, D)$ and $\ell(E, F)$ do not intersect then they are parallel. As a result, if $X \in \ell(C, D) \implies X \notin \ell(E, F)$.

Claim 2 — f is injective.

Proof. $A' = B' \implies d(A', B') = 0 = d(A, B) \implies A = B$. $\square$

As a result, f has an inverse over the range of f , say g . So, $g(A') = A$ for any A in the range of f .

Observe that $d(A', B') = d(g(A'), g(B'))$, as a result, any lemmae that we have proved for f will also work for g .

Hence, if $X' \in \ell(C', D')$ then applying g on both sides and using part (a), we have that $X \in \ell(C, D)$. As a result, $X \in \ell(C, D) \iff X' \in \ell(C', D')$.

So, $X \in \ell(C, D)$ and $X \notin \ell(E, F)$ for all X implies that $X' \in \ell(C', D')$ and $X' \notin \ell(E', F')$ which in turn implies that $\ell(C', D')$ and $\ell(E', F')$ are parallel as well. $\square$